

Research Project Proposal
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INSTRUCTOR:
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MEMORANDUM

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SUBJECT: A Visual Descriptor for Scene Categorization

Enclosed is our group's research project proposal, Scene Categories. This proposal is submitted for partial fulfillment of our project requirement to outline a plan to pursue through the problem formulation with functional requirement, alternative solution generation with different approaches, project management, milestones, task assignments and deliverables. We understand this proposal would undergo a rigorous review assessment, and we are willing to accept any recommendation regarding this matter to modify and resubmit for final project.

Background:

Scene categorization is a fundamental problem in image understanding. It is a challenging task in computer vision and widely applied in many fields, such as image retrieval, video surveillance, travel navigation, medical browser. Large image collections such Internet require efficient organization, that's why in recent years a lot of attempts were performed to design automated systems to solve scene classification task.

Project to be studied:

Scene classification is different from conventional object detection. The scene is a unique composition of several objects related and organized in often unpredictable layout. Those relationships between objects and their organization mostly define the scene class. It makes classification task even more difficult, as a computer is unable to 'understand' such relationships. Object-based judgment is able to provide more or less satisfactory results for man-made scenes as the object functional representation may itself supply sufficient information for labeling a scene. The situation is much more complicated for classifying natural scenes, where separate object detection does not provide enough information to relate the image to a particular category.

Our study aims to obtain high accuracy of scene classification through creating an advanced feature vectors based on multiple feature sets extracted using Gabor filters, CENTRIST and color histograms. The goal is to produce feature vectors that describe the scene as accurately as possible. Each feature extraction method will be first examined alone, then will be combined with features obtained with other methods and tested on accuracy improvement.

Outputs Desired

Classification of natural scenes with 75+% accuracy.

Methodology

Feature extraction:

- Gabor descriptor
- CENTRIST descriptor
- Color histogram descriptor
- Hybrid feature vectors

Classifier:

- Neural Network classifier

GABOR DESCRIPTOR:

Papers:

- [1] Jesmin F. Khan, Reza R. Adhami, Sharif M.A. Bhuiyan, "A customized Gabor filter for unsupervised color image segmentation", Image and Vision Computing, 489-501, 2009
- [2] N. Petkov, P. Kruizinga, "Comparison of texture based features based on Gabor Filters", IEEE Conf. on Image Analysis and Processing, 142-147, 1999

In image processing Gabor descriptors have been used for various tasks including texture segmentation/classification, image recognition and motion tracking. In recent years Gabor filters proved to be robust feature descriptors. Gabor filters possess the optimal localization properties in both spatial and frequency domain. Frequency and orientation representations of Gabor filters are similar to those of the human visual system, and they have been found to be particularly appropriate for texture representation and discrimination. In the spatial domain, a 2D Gabor filter is a Gaussian kernel function modulated by a sinusoidal plane wave.

As it was mentioned before, object recognition is not promising tactic for natural scene classification, but judgement based on textures can be a good approach to classify scenes. Texture can be considered to be repeating patterns of local variation of pixel intensities. Texture analysis is important in many applications of computer image analysis for classification or segmentation of images based on local spatial variations of intensity. Texture describes the structural information of images, so it provides another data to classify comparing to the spectrum. The Gabor filter masks can be considered as orientation and scale tunable edge and line detectors. The statistics of these microfeatures in a given region can be used to characterize the underlying texture information.

Feature extraction using Gabor filters consist of the following steps:

- 1) configuring Gabor filters in the spatio-frequency domain - choosing optimal filter parameters
- 2) combining Gabor filter bank that will be applied on images to form feature vectors

CENTRIST:

Paper: "CENTRIST: A Visual Descriptor for Scene Categorization", Wu and Rehg, 2011

A good descriptor for scene recognition must capture the structure of a given image regardless of its texture, meaning that the stable spatial structure must be captured without taking the details into account. While GIFT is a solution to this requirement, GIFT has very poor performance for indoor scenes.

A new visual descriptor, CENTRIST, is thus created. CENTRIST descriptor encodes the structural properties within an image and is not disturbed by its texture information. CENTRIST (CENSus TRansform hISTogram) descriptor is created by generating the histogram of all the previously computed CT (Census Transform) values of an image. The Census Transform of a pixel is calculated by comparing 3x3 window surrounding pixels to the center pixel and setting vector bits to 1 or 0 depending on the result of the comparison. The cost to compute CENTRIST linear in the number of pixels of the region we are interested in.

The results presented in the paper show that noisy images (digits for example) can still be recognized. This means that the structure is encoded in the CENTRIST descriptor, and thus that the CENTRIST descriptor is very efficient. This is because for 2 different images, even if the intensities

histograms are identical, the CENTRIST descriptors will be very different.

CENTRIST visual descriptor has several advantages comparing to all the other state of the art descriptors: It has a superior recognition performance on multiple standard data sets and has no parameter to tune. Plus, it is very fast (> 50 fps) and easy to implement. The only drawbacks of CENTRIST descriptor is that it is not invariant to rotations or scale changes. Plus, it ignores color information. Note that some drawbacks should not be a problem for scene recognition since images are taken without significant rotation for example.

For images in larger scales, because CENTRIST is not invariant to scale changes, some additional measures have to be taken. Indeed, CENTRIST can only encode global shape structure in a small image patch. We then use a spatial representation based on the Spatial Pyramid Matching scheme (dividing an image into subregions and integrating correspondence results in these regions) for large scale images. It encodes the rough global structure of an image and improves recognition. CENTRIST descriptors of different scales windows are thus concatenated in order to get the final CENTRIST visual descriptors.

COLOR HISTOGRAM:

Papers:

- [1] J. Hafner, H. Sawhney, W. Equitz, M. Flickner and W. Niblack. Efficient color histogram indexing for quadratic form distance functions. PAMI, 1995. 2
- [2] G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. E. Ghaoui and M. I. Jordan. Learning the kernel matrix with semidefinite programming. JMLR, 2004. 4

Bin-ratio provides important information that is going to be used in our project to categorize images. This information is collected from ratios between bin values of histograms that are used for scene and category classifications. Histogram dissimilarity or Bin-Ratio Dissimilarity (BRD) will provide this project with several advantages for category and scene classification tasks.

These advantages are as follows:

- BRD is robust clustering, partial occlusion and histogram normalization.
- BRD captures rich co-occurrence information, and has a linear computational complexity.
- BRD can be easily combined with other dissimilarity measures to gather complementary information for categorization and scene classification tasks in the bag-of-words framework or any other different method.

Histogram-based representation captures rich information that is used in category and scene classification such as in the bag-of-words models, where each image will be represented by the histogram of its quantized visual words to be combined with classifiers or support vector machines for categorization purpose.

Histogram Development and Approaches

- Partial matching information: This is an important issue in category and scene classification because images often contain a large amount of clutter that is irrelevant, and brings noises to the object-of-interest.

- Co-occurrence information: In histogram-based representation co-occurrence captures the correlation information between pairs of histogram bins or joint frequency of visual words in the bag-of-words model.

The above approaches emphasize on three recent matching algorithms with high-order statistics:

- Second-order programming,
- Matching,
- High order matching.

Bin-ratios are defined as the ratios between bin values of histograms. These are robust to normalization, cluttering, and also capture co-occurrence information between all pairs of bins. The results of this are two histograms whose BRD is defined as the sum of squares of normalized differences over all matrix elements. The distances allow cross bin comparison between two histograms to gain robustness to histogram distortions for a better scene categorization.

TIMELINE:

Date	Updates	Details
01/15	FIRST Update Report	Start of 3 descriptors - Gabor: selection of optimal parameters, testing - Centrist: calculation of CT values - Color Histogram descriptor. Image dataset selection.
01/22	SECOND Update Report	First results of 3 descriptors without particular improvements. - Gabor: formation of the Gabor filter bank, - Centrist: calculation of centrist values - Color Histogram descriptor
01/29	THIRD Update Report	First results of 3 descriptors with improvements: - Gabor: feature vector formation, - Centrist - Color Histogram descriptor. Neural Network classifier implementation. Result examination of three approaches individually.
02/05	FOURTH Update Report	Hybrid - combined feature vector of three descriptors. Result examination
02/12	Final Project Report	Comparison of approaches. Conclusion.